Conference

July 8 - 12, 2024
Accademia Pontaniana - Napoli

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Morning session.				
09:30-10:00	Registration	Cattaneo	Mazzocco	Polterovich
10:00-10:30	Opening + Prize			
10:30-11:10	Coffee	Coffee	Coffee	Coffee
11:10-12:00	Zhu	Balibanu	Sheng	Rejzner
12:10-13:00	Matviichuk	Garcia-Fernandez	Poster Buffet	Naef
13:00-15:00				
Afternoon session.				
15:00-15:50	Khesin	Pridham		Bursztyn
16:00-16:40	Coffee	Coffee		Coffee
16:40-17:30	Antonini	Cueca Ten		Schiavina
				Dinner

	Friday
Morning session.	
10:00-10:40	Coffee
10:40-11:30	Fernandes
11:40-12:30	Pym
12:30-14:30	
Afternoon session.	
14:30-15:20	Zhou
15:30-16:10	Coffee
16:10-17:00	Bischoff
17:00-17:30	Weinstein + Closing

Monday, July 8

Chairman: Marco Gualtieri

- 11:10 - 12:00: **Chenchang ZHU** (Universität Göttingen).
  Higher Derived Shifted Symplectic Groupoids.
  In this talk, based on many previous works, we will introduce hopefully a helpful new tool for differential geometers:
  - higher: to deal with quotient singularities
  - derived: to heal transversality problems
  - shifted: for more flexible symplectic situations.

  At the same time of being as complete as possible, we also make the framework as explicit as possible using groupoid models. In the end, we will explain in some concrete examples how this theory is used. This is based on a joint work in progress with Miquel Cueca Ten, Florian Dorsch and Reyer Sjamaar.

- 12:10 - 13:00: **Mykola MATVIICHUK** (Imperial College).
  Deformation Theory of Log Symplectic Manifolds.
  TBA

Chairman: Chiara Esposito

- 15:00 - 15:50: **Boris KHESIN** (University of Toronto).
  Pensive Billiards.
  We define a new class of plane billiards, where while preserving the billiard rule of equality of the angles of incidence and reflection, the billiard ball before reflecting travels along the boundary for some distance depending on the incidence angle. This generalizes so called "puck billiards" proposed by M.Bialy, as well as the motion of a vortex dipole in 2D hydrodynamics near the domain boundary. We prove the variational origin and invariance of a symplectic structure for such billiards. We also observe the appearance of both the golden and silver ratios in the corresponding vortex setting. This is a joint work with D.Glukhovskiy and T.Drivas.

- 16:40 - 17:30: **Paolo Antonini** (Università del Salento).
  Optimal Transport between Algebraic Hypersurfaces.
  I will report on a recent work in collaboration with F. Cavalletti and A. Lerario where we study complex projective hypersurfaces seen as probability measures on the projective space. Our guiding question is: "What is the best way to deform a complex projective hypersurface into another one?" Here the word best means from the point of view of measure theory and mass optimal transportation. In particular we construct an embedding of the space of complex homogeneous polynomials into the probability measures on the projective space and study its intrinsic Wasserstein metric. The Kähler structure of the projective space plays a fundamental role and we combine different techniques from symplectic geometry to the Benamou-Brenier dynamical approach to optimal transportation to prove several interesting facts. Among them we show that the space of hypersurfaces with the Wasserstein metric is complete and geodesic: any two hypersurfaces (possibly singular) are always joined by a minimizing geodesic. Moreover outside the discriminant locus, the metric is induced by a Kähler structure of Weil-Petersson type. In the last part I will give an application to the condition number of polynomial equations solving.

Tuesday, July 9

Chairman: Anton Alekseev

- 09:30 - 10:20: **Alberto CATTANEO** (University of Zurich).
  Poisson Structures from Corners of Field Theories.
  The BV formalism and its shifted versions in field theory have a nice compatibility with boundary structures. Namely, one such structure in the bulk induces a shifted (possibly degenerated) version on its boundary, which can be interpreted as a Poisson structure (up to homotopy). I will present the results for some field theories, in particular, 4D BF theory and 4D gravity.

- 11:10 - 12:00: **Ana BALIBANU** (Louisiana State University).
  Reduction along Strong Dirac Maps.
  We develop a general procedure for reduction along strong Dirac maps, which are a broad generalization of Poisson moment maps. The reduction level in this setting is a submanifold of the target, and the symmetries are given by the action of a groupoid. When applied to group-valued moment maps, this framework recovers several constructions from quasi-Poisson geometry and produces new multiplicative versions of many Poisson varieties that are important to geometric representation theory. This is joint work with Maxence Mayrand.

- 12:10 - 13:00: **Mario GARCIA-FERNANDEZ** (UAM & ICMAT).
  Moduli Spaces and the First Pontryagin Class.
  Principal bundles with vanishing first Pontryagin class play a special role in the theory of higher structures, due to their relation to Courant algebroids and the Chiral de Rham Complex. In this talk I will overview our current knowledge of the geometry of moduli spaces of instantons on principal bundles with vanishing first Pontryagin class, from the moduli space metric to a conjectural Donaldson-Uhlenbeck-Yau type theorem in higher gauge theory.

Chairman: Alessandro Zampini

- 15:00 - 15:50: **Jonathan PRIDHAM** (University of Edinburgh).
  Derived Poisson Structures.
  Shifted Poisson structures have appeared in many guises over the past decades. I will give an overview and examples from the perspective of derived geometry, trying to emphasise similarities with, and differences from, earlier and parallel approaches.

- 16:40 - 17:30: **Miquel CUECA TEN** (Universität Göttingen).
  Transgressions in Poisson Geometry.
  In a broad sense, a transgression map is the composition of a pull-back with a push-forward in a way that some internal degree is changed. A prototypical example is the transgression map from a manifold to its loop space given by pull-back with the evaluation followed by fiber integration. In this talk, I plan to explain the relevance of transgression maps in the construction of secondary characteristic classes for Courant algebroids, n-duals of VB-n-groupoids, integration of Courant algebroids and shifted Lagrangian structures.

Wednesday, July 10

Chairman: Eva Miranda

- 09:30 - 10:20: **Marta MAZZOCCO** (University of Birmingham).
  Triangulations, Cluster Varieties and GDAHA.
  Generalized double affine Hecke algebras (GDAHA) are flat deformations of the group algebras of 2-dimensional crystallographic groups associated to star-shaped simply laced affine Dynkin diagrams. In this talk, we construct embeddings of both GDAHAs of type D_4 and E_6 into matrix algebras over quantum cluster χ-varieties, thus linking to the theory of higher Teichmüller spaces.

- 11:10 - 12:00: **Yunhe SHENG** (Jilin University).
  Post-groupoids and the Yang-Baxter Equation.
  In this talk, first we recall the notion of a post-group. By differentiation of a post-Lie group, one can obtain a post-Lie algebra, which was introduced by Vallette in 2007 and have important applications in numerical integration on manifolds and Martin Hairer's regularity structures. There are close relationships between post-groups, Rota-Baxter groups, skew-left braces and Lie-Butcher groups. In particular, post-groups give rise to matched pairs of groups, and can be used to construct set-theoretical solutions of the Yang-Baxter equation. Then we introduce the notion of a post-groupoid. A post-groupoid is a group bundle equipped with another binary operation. The section space of a post-groupoid is a weak post group. A post-groupoid gives rise to a groupoid and an action on the original group bundle. By differentiation of a post-Lie groupoid, one can obtain a post-Lie algebroid, which was introduced by Munthe-Kaas and Lundervold in the study of geometric numerical analysis. Finally, we show that post-groupoids provide solutions of the Yang-Baxter equation on quivers. This is a joint work with Rong Tang and Chenchang Zhu.

Thursday, July 11

Chairman: Pavol Severa

- 09:30 - 10:20: **Leonid POLTEROVICH** (Tel Aviv University).
  Big Fiber Theorems, Ideal-Valued Measures, and Symplectic Topology.
  I will discuss an adaptation of Gromov's ideal-valued measures to symplectic topology. It leads to a unified viewpoint at three "big fiber theorems": the Centerpoint Theorem in combinatorial geometry, the Maximal Fiber Inequality in topology, and the Non-displaceable Fiber Theorem for involutive maps in symplectic topology, and yields applications to symplectic rigidity. Joint work with Adi Dickstein, Yaniv Ganor, and Frol Zapolsky.

- 11:10 - 12:00: **Kasia REJZNER** (University of York).
  Quantization of Gauge Theories in Perturbative Algebraic Quantum Field Theory.
  I will begin with outlining the formalism of perturbative algebraic quantum field theory, which has been successfully applied in quantization of gauge theories. Then I will explain how it can be generalized to situations where a boundary is present, including the case of an asymptotic "boundary at infinity." The key idea is to start with a Poisson structure and quantize it through formal deformation quantization. In the presence of boundaries and corners, one has a whole hierarchy of such structures.

- 12:10 - 13:00: **Florian NAEF** (Trinity College Dublin).
  Simple Homotopy Types in String Topology.
  Reidemeister and Whitehead gave a completely algebraic description of finite CW complexes up to cell collapses (aka simple homotopy types). We will see how a weakening of this structure (namely its trace) enters into the construction of a string topology operation on the homology of the free loop space (the loop coproduct). This in particular implies that string topology is not a homotopy invariant. This is joint work with Pavel Safronov.

Chairman: Marco Zambon

- 15:00 - 15:50: **Henrique BURSZTYN** (IMPA).
  Symplectic Groupoids via 2-Shifted Lagrangian Structures.
  I will discuss the role of 2-shifted Lagrangian structures as a framework for constructing (quasi-) symplectic groupoids of interest in Poisson geometry. I will focus on Lagrangian structures on Lie groupoid morphisms into (2-shifted symplectic) Lie groups and explain their infinitesimal counterparts in terms of Dirac structures. Applications include the description of integrations of quasi-Poisson spaces and the construction of symplectic groupoids of Poisson homogeneous spaces. The talk is based on joint work with D. Alvarez and M. Cueca.

- 16:40 - 17:30: **Michele SCHIAVINA** (Università di Pavia).
  Reduction by Stages for Gauge Theory with Corners.
  I will discuss the Hamiltonian formulation of local field theories on manifolds with corners in the particular, yet common, case in which they admit an equivariant momentum map. In the presence of corners, the momentum map naturally splits into a part encoding "Cauchy data" or constraints, and a part encoding the "flux" across the corner.

  The reduced phase space is then conveniently handled via symplectic reduction by stages, adapted to local group actions.

  When smooth, the output of this analysis are natural Poisson structures associated to corner submanifolds (both on- and off-shell), leading to the concept of (classical) flux superselection sectors as their symplectic leaves, and providing a roadmap to quantum superselection. Moreover, in terms of higher structures, embedding this "on shell" data in ambient corner configuration spaces leads to "Poisson-infinity data".

  As a byproduct, both the Peter-Weyl theorem and the Verlinde formula are seen as descending from the superselection factorisation for the appropriate gauge theory. Looking at electrodynamics or gravitation, other factorisation formulas are expected, owing to the general structure of reduced phase spaces.

Friday, July 12

Chairman: Antonio De Nicola

- 10:40 - 11:30: **Rui Loja FERNANDES** (University of Illinois Urbana-Champaign).
  Proper Symplectic Groupoids.
  In this talk, I will provide a survey of the rich geometry underlying proper symplectic Lie groupoids. This is based on ongoing joint work with Marius Crainic (Utrecht) and David Martinez Torres (Madrid).

- 11:40 - 12:30: **Brent PYM** (McGill University).
  Semi-classical Hodge Theory.
  Kontsevich's deformation quantization formula associates to any Poisson manifold an algebra of "quantum observables", defined as a noncommutative deformation of the product of functions. The formula is Feynman-style series expansion, whose coefficients are multiple zeta values, making it intractable for direct calculation. Following a suggestion of Kontsevich, I will explain how K-theory and mixed Hodge structures can be used to construct natural "period coordinates" on the moduli space of smooth Poisson varieties, in which the quantization can often be computed simply, explicitly and nonperturbatively as the exponential map for a complex torus. This gives a conceptual explanation for the appearance of various classical transcendental functions in the relations defining well-known noncommutative algebras. This talk is based on joint work with A. Lindberg.

Chairman: Luca Vitagliano

- 14:30 - 15:20: **Yan ZHOU** (Northeastern University).
  Stokes Geometry and WKB Analysis.
  We study the Riemann-Hilbert maps of a class of irregular connections on the trivial rank n bundle over the projective line that appears in various geometry and representation theory contexts. The Stokes data give the generalized monodromy data of these connections. The compactification of the parameter space of the irregular types of these connections is $\bar{M}_{0,n}$. At the deepest stratum of $\bar{M}_{0,n}$, we show that the WKB leading terms of the Stokes matrices are periods on the spectral curves. Near the deepest stratum of $\bar{M}_{0,n}$, we study the WKB

approximation of Stokes data via the frame of spectral networks. In particular, we will explain how interlacing inequalities indicated by Poisson geometry correspond to BPS states. If time permits, we will comment on the picture of the quantized Riemann Hilbert maps. The talk will be based on joint work with Alekseev, Neitzke, and Xu.

■ 16:10 - 17:00: **Francis BISCHOFF** (University of Regina).
b^k Poisson Geometry and the Character Variety.
The b^k tangent bundle, first introduced by Scott, is a Lie algebroid consisting of vector fields tangent to a hypersurface D to order k. Although this algebroid depends on the choice of a local defining function for D, all functions give rise to isotopic Lie algebroids. In this talk I will introduce a wider class of Lie algebroids that are locally of b^k type but which are classified, up to isotopy, by a local system on D. These algebroids allow us to define a new class of Poisson structures, which are symplectic away from D, and whose geometry is controlled by a local system on D. I will give examples of these Poisson structures obtained by deformations along paths in a character variety. This is joint work with Álvaro Del Pino and Aldo Witte.

Posters

- **Lory Aintablian** (MPIM Bonn).
  Differentiation of elastic diffeological groupoids.
  I will present a differentiation procedure from elastic diffeological groupoids to generalized Lie algebroids in the category of diffeological spaces. First, I will show the differentiation procedure of groupoid objects in any category with an abstract tangent structure in the sense of Rosický. Since the left Kan extension of the tangent structure of smooth manifolds defines a tangent structure on elastic diffeological spaces, this construction applies to elastic diffeological groupoids. This is joint work with Christian Blohmann.

- **Daniel Alvarez** (University of Toronto).
  Transitive Courant algebroids and double symplectic groupoids.

- **Camilo Angulo** (Jilin University).
  Van Est theory for strict Lie 2-groups.
  As it has been proven lately, the homological perturbation lemma can be used to give a conceptual explanation for the van Est map and a clear method from which van Est theorem follows. In this poster, we present the double complex that the perturbation lemma should be applied to in the case of strict Lie 2-groups; we explain how this can be regarded as a differentiation method; we present the corresponding van Est map and theorem; and, as an application, present the differentiation of the Segal 2-form on the loop group.

- **Eugenia Boffo** (Comenius University Bratislava).
  BV and pictures of the spinning particle.
  Similarly to the Ramond-Neveu-Schwarz superstring, a spinning particle is a relativistic particle with a supersymmetric worldline. After Becchi-Rouet-Stora-Tyutin quantization, different sectors arise for the representation of the bosonic ghosts associated with supersymmetry and are distinguished by different picture numbers [2]. Though the maximal picture case (integral forms) and the minimal one (superforms) are cohomologically equivalent, a different situation emerges

for the intermediate case, that of pseudoforms. In our recent work [1] we have computed the partition function in all of the pictures and highlighted the presence of topological sectors for the case of pseudoforms. In the same fashion as second quantizing a string leads to string field theory, for a spinning particle second quantization yields a (gauge) field theory. Motivated by the work of Zwiebach [3], which shows the presence of a homotopy algebraic structure (the Batalin-Vilkovisky one) in quantized closed string field theory, one may wonder if the same structure is present in the case of a particle. After all, BRST itself recovers the field theory up to linear order (or otherwise stated, the BRST operator is the first product in the expansion in multiproducts). We suggest [1] that indeed one can recover a BV formulation of Yang-Mills theory (as target space theory for the spinning particle).This possibility relies on the existence of a convenient deformation of the BRST operator by fields.

[1] E. Boffo et al. In: (2024). arXiv: 2402.09868 [hep-th].

[2] E. Witten. In: *Pure Appl. Math. Quart.* 15 (2019).

[3] B. Zwiebach. In: *Nucl. Phys. B* 390 (1993).

■ **Pedro Henrique Carvalho Silva** (IMPA Rio de Janeiro).
Homological reduction of Poisson and Courant structures.

■ **Ioana Ciuclea** (West University of Timisoara).
Dual pairs and coadjoint orbits of nonlinear flags.
We present a more general version of the ideal fluid dual pair, involving the group of diffeomorphisms preserving a weighted nonlinear flag. Symplectic reduction for the right leg yields coadjoint orbits of the Hamiltonian group, that consist of weighted nonlinear flags.

■ **Alessandro Pietro Contini** (Leibniz Universität Hannover).
Symplectic geometry on scattering manifolds.
Symplectic geometry on singular manifolds, a topic that has received considerable attention in the Poisson community, originated in an analytical framework in the work of Melrose on Fourier Integral Operators on manifolds with boundary. Recently, scattering-symplectic

geometry has been introduced by Lanius to study higher-order singularities. In this poster, I want to present a notion of symplectic structure and symplectomorphisms on scattering manifold, differing slightly from this previous work, that is however very rooted in the original approach and should lead to a good analytical theory

- **Oscar Cosserat** (Georg-August Universität Göttingen).
  Post-Lie algebras and Poisson integrators.

- **Sourav Das** (Chennai Mathematical Institute).
  Moduli stack of Higgs bundles on stable curves.
  In this article, we construct a flat degeneration of the derived moduli stack of Higgs bundles on smooth curves using the stack of expanded degenerations of Jun Li. We show that there is an intrinsic relative log-symplectic form on the degeneration and we compare it with the one constructed by the second author. We show that the Hitchin map of the degeneration we construct has complete fibers. Furthermore, we show that the Hitchin map is flat and that a suitable open subset of the smooth locus of the reduced nilpotent cone is Lagrangian. We also extend the construction of the moduli of Higgs bundles along with the relative log-symplectic form over the universal moduli stack of stable curves.

- **Thomas De Fraja** (Heriot–Watt University).
  Courant algebroid relations and T-duality.

- **Lennart Döppenschmitt** (University of Zurich).
  Kähler geometry of brane moduli spaces.

- **Ilias Ermeidis** (Georg-August Universität Göttingen).
  On Deformations of Infinitesimally Multiplicative Distributions on Lie Algebroids.
  In this poster, we present several features of the deformation theory of IM-distributions, as well ideals, inside Lie algebroids.

■ **Benedetta Facciotti** (University of Birmingham).
An example of Stokes data and real decorated blowups.
This poster focuses on the study of a specific example of a meromorphic connection on a vector bundle of rank 3 on the projective line. It will be shown how the Stokes data arising from the higher order pole can be computed exploiting the decorated real oriented blowup of the surface involved and Fock-Goncharov theory.

■ **Alfonso Garmendia** (CRM).
Properties of E-symplectic Manifolds.

■ **Iván Gutiérrez Sagredo** (Universidad de Burgos).
Unimodularity and invariant volume forms for hamiltonian dynamics on Poisson-Lie groups and Poisson homogeneous spaces.
In this poster, we discuss several relations between the existence of invariant volume forms for Hamiltonian systems on Poisson-Lie groups and Poisson homogeneous spaces and the unimodularity of the Poisson-Lie structure. Some of the results presented are based on [1].

[1] I Gutierrez-Sagredo et al. "Unimodularity and invariant volume forms for Hamiltonian dynamics on Poisson–Lie groups". In: *Journal of Physics A: Mathematical and Theoretical* 56 (2023).

■ **Yuji Hirota** (Azabu University).
Symmetry in Bundle-Valued n-plectic Geometry.

■ **Douwe Hoekstra** (Radboud University).
Poisson cohomology of three-dimensional linear Poisson structures.
Linear Poisson structures are in one-to-one correspondence with Lie algebras. This allows us to classify all linear Poisson structures in three dimensions up to isomorphism. Using this classification, originally due to Bianchi, we compute the Poisson cohomology groups of all linear Poisson structures in three dimensions. This work continues

on the work of A. Weinstein & V. Ginzburg for so(3) and F. Zeiser & I. Marcut for sl(2, R). This is joint work with F. Zeiser.

■ **Gavin Hurd** (University of Toronto).
PREQUANTIZATION OF AN ODD EXACT COURANT ALGEBROID.

■ **Naoki Kimura** (Tokyo University of Science).
COMPATIBILITY BETWEEN JACOBI STRUCTURES AND PSEUDO -RIEMANNIAN COMETRICS ON JACOBI ALGEBROIDS.
Boucetta introduced compatibility between Poisson structures and pseudo-Riemannian cometrics on manifolds. In this poster presentation, we define compatibility between Jacobi structures and pseudo-Riemannian cometrics on Jacobi algebroids, as a generalization of the compatibility defined by Boucetta. We also state that this compatibility with a cometric is "preserved" by the Poissonization of a Jacobi structure. In particular, for a contact metric structure on a manifold, satisfying the compatibility condition is equivalent to being a Sasakian structure. This research is a joint work with Tomoya Nakamura (Kogakuin University).

■ **Charlotte Kirchhoff-Lukat** (KU Leuven).
LOG SYMPLECTIC SURFACES AND WRAPPED FUKAYA CATEGORIES.
In this talk I will explain how to apply the techniques of partially wrapped Fukaya categories and stop removal (due to Ganatra-Pardon-Shende) to log symplectic surfaces. Every closed log symplectic surface can be decomposed into open symplectic surfaces and tubular neighbourhoods of the log degeneracy loci. Having already directly constructed the Fukaya category of any whole closed log symplectic surface, we can use this decomposition as a tool for explicit computation: While computing the whole category explicitly quickly becomes complicated for surfaces of higher genus and with more complicated degeneracy loci, the wrapped Fukaya categories of the smaller components are simpler to compute and can then be combined to obtain the Fukaya category of the whole.

■ **Praphulla Koushik** (NIT Calicut).
On certain vector bundles over Dirac manifolds.
Motivated by the study of Poisson vector bundles over Poisson manifolds by Viktor Ginzburg in their work "Grothendieck Groups of Poisson Vector Bundles",we introduce the notion of a certain vector bundle over Dirac manifolds and study certain properties. This will be based on our work in progress.

■ **Kalin Krishna** (Georg–August Universität Göttingen).
Higher Lie Groupoids and Bibundles.
We define the notion of Higher groupoids in Good Geometric Categories and organise them into an $(\infty, 1)$-category framework, where the morphism between the higher groupoids are given by bibundles between the groupoids. One particular example is the bibundle version of higher morphisms between higher Lie groupoids as Kan fibrations over interval and higher simplices. This approach gives a more combinatorial and geometric way of approaching anafunctors and higher generalised morphisms between Lie groupoids.

■ **Andrey Krutov** (Charles University).
Clifford algebra analogue of the Cartan theorem.
Let $\mathfrak{g}$ be a complex simple Lie algebra. The Hopf-Koszul-Samelson theorem asserts that the algebra of $\mathfrak{g}$-invariants in the exterior algebra of $\mathfrak{g}$ is the exterior algebra over the space of primitive invariants P. Kostant proved the analogous result for the Clifford algebra of $\mathfrak{g}$. Namely, the algebra of $\mathfrak{g}$-invariants in $Cl(\mathfrak{g})$ is the Clifford algebra over the space of primitive invariants. Let $\mathfrak{k}$ be a symmetric subalgebra of $\mathfrak{g}$ and $\mathfrak{p}$ be the corresponding isotropy representation. The Cartan theorem states that the algebra of $\mathfrak{k}$-invariants in the exterior algebra of $\mathfrak{p}$ is isomorphic to the tensor product of the exterior algebra of the Sameson subspace of P, corresponding to the pair $(\mathfrak{g}, \mathfrak{k})$, and a certain commutative algebra A. We prove a Clifford algebra analogue of the Cartan theorem. Namely, we show that the algebra of $\mathfrak{k}$-invariants in $Cl(\mathfrak{p})$ is the tensor product of the Clifford algebra over the space of primitive invariant of $\mathfrak{p}$ with a certain filtered deformation of A.

This is joint work with K. Calvert, K. Grizelj, and P. Pandžić.

■ **Joshua Lackman** (Peking University).
A Groupoid Construction of Functional Integrals.

■ **Aidan Lindberg** (University of Toronto).
Picard Groups of Holomorphic Poisson Manifolds.

■ **Ruben Louis** (Jilin University).
On Nash-type resolution of Lie algebroids with applications to (twisted) Poisson structures.
We introduce a Nash-type blowup method on (almost) Lie algebroids with applications, particularly in the context of (twisted) Poisson structures. It also makes sense on singular subalgebroids in the sense of Androulidakis-Zambon. The study reveals that any Lie algebroid transforms under this blowup to a new Lie algebroid whose anchor map is injective on an open dense subset.

■ **Ana Carolina Mançur** (IMPA Rio de Janeiro).
Manin triples for Poisson double structures.
Manin triples offer an alternative perspective to Lie bialgebras (or Lie bialgebroids), which are infinitesimal counterparts of Poisson groups (or groupoids). In this poster, we will discuss Manin triples in the context of double structures that arise as infinitesimal versions of double Poisson groupoids. We will observe that Drinfeld doubles in this context yield structures known as LA-Courant algebroids and Multiplicative Courant algebroids, indicating the relationship between them through differentiation and integration.

■ **Filip Moučka** (Universitat Autònoma de Barcelona).
Symmetric Poisson structures.
A Poisson structure on a smooth manifold is a skew-symmetric bivector field satisfying a certain integrability condition. We give an answer to the very natural and simple question: What is a suitable integrability condition for a symmetric bivector field? The definition and basic properties of so-called symmetric Poisson structures are presented.

■ **Joshua Mundinger** (University of Wisconsin-Madison).
THE HOCHSCHILD-KOSTANT-ROSENBERG FILTRATION IN CHARACTERISTIC P.
In characteristic zero, the HKR isomorphism describes Hochschild homology of a variety in terms of differential forms. In characteristic p, we describe the first possibly nonzero obstruction to this isomorphism using deformation quantization.

■ **Tomoya Nakamura** (Kogakuin University).
DIRAC PAIRS ON JACOBI ALGEBROIDS.
Kosmann-Schwarzbach introduced Dirac pairs on Lie algebroids. In this talk, we introduced Dirac pairs on Jacobi algebroids. these are a generalization of Dirac pairs on Lie algebroids. We show the relationship between Dirac pairs on Lie and on Jacobi algebroids, and that Dirac pairs on Jacobi algebroids characterize several compatible structures on Jacobi algebroids. For example, JΩ-structures and ΩN-structures.

■ **Clarice Netto** (University of São Paulo).
ALGEBRAIC ASPECTS OF MULTIPLICATIVE SECTIONS OF CA-GROUPOIDS.
Motivated by recent results of C. Ortiz and J. Waldron about the category of multiplicative sections of VB-groupoids and LA-groupoids, we study the algebraic structure underlying the category of multiplicative sections of a multiplicative Courant algebroids (CA-groupoids), which we obtain a Courant 2-algebra. We also discuss quasi-isomorphisms of the Courant 2-algebras between CA-groupoids Morita equivalents. This poster is based on a joint work in development with C. Ortiz and E. Cupitra.

■ **Nikita Nikolaev** (University of Birmingham).
POISSON GEOMETRY OF THE HODGE MODULI SPACE.
The Hodge moduli space parameterises the data $(X, D, \mathcal{E}, \nabla)$ consisting of compact complex curves X with effective divisors D, and holomorphic vector bundles $\mathcal{E}$ with meromorphic λ-connections ∇ with poles bounded by D. This space carries a complex hyperkahler structure induced by the isomonodromy connection. It also carries an integrable system structure over the full Hitchin base with Lagrangian

fibres being algebraic tori $(\mathbb{C}^\times)^{2d}$ which are the $\mathbb{C}^\times$-character varieties of spectral curves. This is joint work in progress with Tom Bridgeland.

■ **Taika Okuda** (Tokyo University of Science).
Explicit Formula for Star Product with Separation of Variables of Two-dimensional Locally Symmetric Kähler Manifolds.
We explain the explicit formula that gives the star product with separation of variables of two-dimensional locally symmetric Kähler manifolds. The construction methods of deformation quantization with separation of variables were proposed by Sako-Suzuki-Umetsu and Hara-Sako for locally symmetric Kähler manifolds. These methods state that for any locally symmetric Kähler manifolds, there exists the star product that is concretely given from the solution of some recurrence relations. From their construction method, the explicit formula for a star product on arbitrary ones was obtained only in one-dimensional case. In this work, we give the explicit formula for a star product on two-dimensional locally symmetric Kähler manifolds with matrix representations from the construction method proposed by Hara-Sako. By calculating the star products of $\mathbb{C}^2$ and $\mathbb{C}P^2$ as concrete examples from the obtained explicit formula, we also confirm that their star products coincide with the results obtained in previous works. This is the joint work with Akifumi Sako (Tokyo University of Science).

■ **Stefano Ronchi** (Georg-August Universität Göttingen).
Higher cotangent groupoids.

■ **Philipp Schmitt** (Leibniz Universität Hannover).
Strict quantization of polynomial Poisson structures.

■ **Wilmer Smilde** (University of Illinois Urbana–Champaign).
Lie theory & classification of geometric structures: Bryant algebroids.
Already implicit in the work of Elie Cartan, there is a deep connection between classification problems for geometric structures (coframes) and modern Lie theory. Many classification problems arise as a 'PDE in the structure functions of a coframe'. In rare (but not unnatural) cases, this PDE is of so-called finite type, and defines a Lie algebroid that classifies the problem. The problem of classifying all complete (global) solutions is essentially the same as determining whether the Lie algebroid is integrable. This deep connection between Lie theory and geometric classification problems has been worked out by Fernandes and Struchiner.

We propose a generalization of Lie algebroids, a Bryant algebroid, that is designed to capture the essential structure of the 'infinite type PDEs in the structure functions of a coframe'. The theory of Bryant algebroids makes the connection between generic classification problems for geometric structures and Lie theory clear and precise. This is ongoing work with Rui Loja Fernandes.

■ **Rudolf Smolka** (Czech Technical University in Prague).
Geometric Approach to Graded Vector Bundles.
In the theory of $\mathbb{Z}$-graded manifolds one usually defines the so-called graded vector bundles first by their (graded) sheaf of sections, and then proceeds to construct the underlying graded manifold. In this poster we show a "geometric" definition much more akin to the usual approach to vector bundles in ordinary (non-graded) differential geometry. Some examples and properties of graded vector bundles are also presented.

■ **Andreas Swerdlow** (University of Manchester).
A Graphical Calculus for Microformal Morphisms and Pullbacks.
Microformal or thick morphisms, introduced by Ted Voronov, are a generalisation of smooth maps between manifolds that still give rise to pullbacks on functions. These pullbacks are in general nonlinear and formal, and in special cases they define L-infinity morphisms be-

tween the algebras of functions on homotopy Poisson or homotopy Schouten manifolds. My poster will display my recent work developing a graphical calculus which calculates all terms in the formal power series that result from these pullbacks. The method heavily resembles the use of Feynman diagrams in perturbative quantum field theory, and the relationship between the calculi for classical and quantum thick morphisms is exactly the relationship between the tree-level and full perturbative expansions in QFT.

■ **Fabricio Valencia** (University of São Paulo).
CLOSED GEODESICS ON RIEMANNIAN STACKS.
We present a result which shows the existence of closed geodesics of positive length on certain separated Riemannian stacks, thus generalizing some of the well known results for the case of orbifolds due to Guruprasad-Haefliger. Our approach follows two main steps. Firstly, motivated by the case of orbifolds, we proceed by extending classical techniques from infinite dimensional Morse theory applied to the stacky energy functional over the space of stacky loops in a Riemannian stack that is presented by a proper regular groupoid. Secondly, we use the regularization result for separated Riemannian stacks due to Posthuma-Tang-Wang in order to push stacky geodesics of positive length down. Based on joint work with C. Ortiz and L. Vitagliano.

■ **Mikhail Vasilev** (University of Glasgow).
RUIJSENAARS DUALITY FOR B, C, D TODA SYSTEMS.
We use theHamiltonian reduction method to construct the Ruijsenars dual systems to generalized Toda chains associated with the classical Lie algebras of types (B, C, D). The dual systems turn out to be the B, C, and D analogues of the rational Goldfish model, which is, as in the type A case, the strong coupling limit of rational Ruijsenaars systems. We explain how both types of systems emerge in the reduction of the cotangent bundle of a Lie group and provide the formulae for dual Hamiltonians and action-angle duality maps. We compute explicitly the higher Hamiltonians of Goldfish models using Cauchy–Binet theorem.

■ **Dan Wang** (CUHK).
Geometric Quantization of Kähler Manifold with T-Symmetry.

■ **Aldo Witte** (Antwerp University).
Jets of foliations and Lie algebroids.
In this talk we will study a higher version jet version of the Riemann-Hilbert correspon- dence. The classical Riemann-Hilbert correspondence relates linear representations of the fundamental group of manifold with flat vector bundles over the manifold. Instead of linear representations we will study representations which are polynomial in nature. This leads to the discovery of the notion of a k-jet of foliation, a distribution which is only involutive up to order k. We will then show that these k-jets of foliations can be used to classify a certain group of Lie algebroids arising in singular symplectic geometry. Joint work with Alvaro del Pino and Francis Bischoff.

■ **Xiaomeng Xu** (Peking University).
Quantum Stokes matrices.

■ **Fabrizio Zanello** (University of Potsdam).
Second order two-fields variational characterization of $(2 + 1)$-dimensional modified sixth-order Boussinesq-type equations and extensions of Lagrangian equivalents.
We determine second order Lagrangians for $(2+1)$-dimensional modified Boussinesq equations and we discuss some aspects concerning conservation laws and conserved quantities associated with invariance properties of their extended 'full' equivalents, in particular of Krupka–Betounes type. (Joint work with Frabrizio Zanello)